supervised → labeled data $\{x^{(i)}, y^{(i)}\}$

unsupervised → unlabeled data $\{x^{(i)}\}$

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k-Means

cluster centroid

Distortion $J(c, \mu) = \sum_i \|x^{(i)} - \mu_{c^{(i)}}\|^2$

Note.

J는 단조감소

⇒ J는 수렴 run multiple times

(i) starts with $\{x^{(1)}, \dots, x^{(m)}\}$ $\{\mu^{(1)}, \dots, \mu^{(k)}\}$

data centroids

(ii) update $c^{(i)}$ (계산중) : 가장 가까운 centroid를 가진 클러스터에 할당

(iii) update $\mu^{(j)}$ (갱신) : cluster의 속하는 점들을 다시 계산.

Mixture of Gaussian Model

- data $\{x^{(1)}, \dots, x^{(m)}\}$
- $p(x^{(i)}, z^{(i)}) = p(x^{(i)} | z^{(i)}) \, p(z^{(i)})$

  $\downarrow$ $\mathcal{N}(\mu, \Sigma)$ $\quad$ $\downarrow$ Multinomial(ϕ)

(i) choose $z^{(i)}$

(ii) choose $x^{(i)}$ from Gaussian

- likelyhood $= \sum_i \log p(x^{(i)})$

  $= \sum_i \log \sum_{z^{(i)}} p(x^{(i)} | z^{(i)}) \, p(z^{(i)})$

maximize

(i) $\phi_j = \frac{n_j}{n}$

(ii) $\mu_j = \frac{1}{n_j} \sum_{i \in c_j} x^{(i)}$

(iii) $\Sigma_j = \frac{1}{n_j} \sum_{i \in c_j} (x^{(i)} - \mu_j)(x^{(i)} - \mu_j)^T$

- if we knew $z^{(i)}$

  then likelyhood $= \sum_i \log p(x^{(i)} | z^{(i)}) + \log p(z^{(i)})$

similar to GDA

EM Algorithm

- E-step : guess $z^{(i)}$ i.e $w_j^{(i)} = p(z^{(i)} = j \mid x^{(i)})$

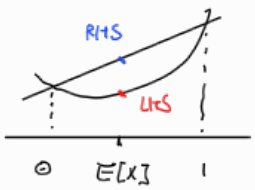
$$= \frac{p(x^{(i)} | z^{(i)} = j) \, p(z^{(i)} = j)}{\sum_l p(x^{(i)} | z^{(i)} = l) \, p(z^{(i)} = l)}$$

- M-step : update params ϕ_j, μ_j, Σ_j

Jensen's Inequality $f(E[X]) \leq E[f(X)]$

(i) convex f $\Rightarrow$

(ii) random variable X



Note.

f strictly convex

then $f(E[X]) < E[f(X)]$

- density $p(x) = \sum_z p(x, z)$
- log-likelyhood of the data $\ell(\theta) = \sum_i \log p(x^{(i)}) = \sum_i \log \sum_z p(x^{(i)}, z)$

Note.

suppose each $z^{(i)}$ observed

otherwise it is non-convex optimization problem : difficult.

- $\sum_z Q(z) = 1$, $Q(z) \geq 0$ (distribution)
- $\log p(x) = \log \sum_z p(x, z)$

$$= \log \sum_z Q(z) \frac{p(x, z)}{Q(z)} \overset{\text{Jensen}}{\geq} \sum_z Q(z) \log \frac{p(x, z)}{Q(z)} = \text{ELBO}$$

---

if you want this to be equality : $\frac{p(x, z)}{Q(z)} = \text{constant}$

- $Q(z) = p(z|x)$ i.e. $Q(z) \propto p(x, z)$

$$= \frac{p(x, z)}{p(x)} = \frac{p(x, z)}{\sum_z p(x, z)}$$

- E-Step : $Q_i(z^{(i)}) = p(z^{(i)} | x^{(i)}; \theta)$
- M-Step : $\theta = \arg\max_\theta \sum_i \text{ELBO}(x^{(i)}; Q_i, \theta)$

$$= \arg\max_\theta \sum_i \sum_{z^{(i)}} Q_i(z^{(i)}) \log \frac{p(x^{(i)}, z^{(i)}; \theta)}{Q_i(z^{(i)})}$$

Note.

Maximize $\text{ELBO}(Q, \theta)$

in respect to Q (E) θ (M)

- $\ell(\theta^{(t+1)}) \geq \sum_{i=1}^n \text{ELBO}(x^{(i)}; Q_i^{(t)}, \theta^{(t+1)})$

$$\geq \sum_{i=1}^n \text{ELBO}(x^{(i)}; Q_i^{(t)}, \theta^{(t)}) = \ell(\theta^{(t)})$$

therefore algorithm converges

• $ELBO(x) = \mathbb{E}_{z\sim Q}[\log p(x,z)] - \mathbb{E}_{z\sim Q}[\log Q(z)]$

$= \mathbb{E}_{z\sim Q}[\log p(x|z)] - D_{KL}(Q\|p_z)$ with $D_{KL}(Q\|p_z) = \sum_z Q(z)\log\frac{Q(z)}{p(z)}$

$\mathbb{E}_{z\sim Q}[\log p(x|z)]$: M step에서 업데이트.

$D_{KL}(Q\|p_z)$: E step에서 업데이트.

$= \log p(x) - D_{KL}(Q\|p_{z|x})$

$Q = p_{z|x}$ 일 때 최적.

• E step : $w_j^{(i)} = Q_i(z^{(i)}=j) = p(z^{(i)}=j \mid x^{(i)})$

• M step : $ELBO = \sum_i \sum_{z^{(i)}} Q_i(z^{(i)}) \log \frac{p(x^{(i)}, z^{(i)})}{Q_i(z^{(i)})}$

$= \sum_i \sum_j Q_i(z^{(i)}=j) \log \frac{p(x^{(i)}|z^{(i)}=j)\,p(z^{(i)}=j)}{Q_i(z^{(i)}=j)}$

$= \sum_i \sum_j w_j^{(i)} \log \frac{\frac{1}{(2\pi)^{d/2}|\Sigma_j|^{1/2}} \exp\left(-\frac{1}{2}(x^{(i)}-\mu_j)^T \Sigma_j^{-1}(x^{(i)}-\mu_j)\right)\cdot \phi_j}{w_j^{(i)}}$

$\nabla_{\mu_l}(ELBO) = -\nabla_{\mu_l} \sum_i \sum_j w_j^{(i)} \frac{1}{2}(x^{(i)}-\mu_j)^T \Sigma_j^{-1}(x^{(i)}-\mu_j)$

$= \frac{1}{2}\sum_i w_l^{(i)} \nabla_{\mu_l}\left(2\mu_l^T \Sigma_l^{-1} x^{(i)} - \mu_l^T \Sigma_l^{-1}\mu_l\right)$

$= \sum_i w_l^{(i)}\left(\Sigma_l^{-1}x^{(i)} - \Sigma_l^{-1}\mu_l\right) = 0$ therefore $\mu_l = \frac{\sum_i w_l^{(i)} x^{(i)}}{\sum_i w_l^{(i)}}$

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respect to ϕ_j

(i) maximize $\sum_i \sum_j w_j^{(i)} \log \phi_j$

(ii) $\sum_j \phi_j = 1$

$\Rightarrow \mathcal{L}(\phi) = \sum_i \sum_j w_j^{(i)} \log \phi_j + \beta\left(\sum_j \phi_j - 1\right)$

$\sum_j \phi_j = 1$ implies $-\beta = n$

$\Rightarrow \frac{\partial \mathcal{L}}{\partial \phi_j} = \sum_i \frac{w_j^{(i)}}{\phi_j} + \beta = 0$ therefore $\phi_j = \frac{\sum_i w_j^{(i)}}{-\beta} = \frac{1}{n}\sum_i w_j^{(i)}$